

Formulaire des transformées de Fourier usuelles

$f(t)$	$\hat{f}(\omega)$
$\rho_a(t) = \begin{cases} 1 & \text{si } t \leq \frac{a}{2} \\ 0 & \text{sinon} \end{cases} \quad (a > 0)$	$\begin{cases} \frac{2 \sin(a \frac{\omega}{2})}{a \omega} & \text{si } \omega \neq 0 \\ \text{sinon} & \end{cases}$
$\frac{\sin at}{t} \quad (a > 0)$	$\pi \rho_{2a}(\omega)$
$q_a(t) = \begin{cases} 1 - \frac{ t }{a} & \text{si } t \leq a \\ 0 & \text{sinon} \end{cases} \quad (a > 0)$	$\begin{cases} \frac{4 \sin^2(a \frac{\omega}{2})}{a \omega^2} & \text{si } \omega \neq 0 \\ \text{sinon} & \end{cases}$
$\frac{\sin^2 at}{t^2} \quad (a > 0)$	$\pi a q_{2a}(\omega)$
$e^{-a t } \quad (a > 0)$	$\frac{2a}{a^2 + \omega^2}$
$\frac{1}{a^2 + t^2} \quad (a > 0)$	$\frac{\pi}{a} e^{-a \omega }$
$e^{-at} \mathcal{U}(t) \quad (a \in \mathbb{C}, \Re(a) > 0)$	$\frac{1}{a + i\omega}$
$te^{-at} \mathcal{U}(t) \quad (a \in \mathbb{C}, \Re(a) > 0)$	$\frac{1}{(a + i\omega)^2}$
$e^{-at} \sin(bt) \mathcal{U}(t) \quad (a \in \mathbb{C}, \Re(a) > 0, b \in \mathbb{R})$	$\frac{b}{(a + i\omega)^2 + b^2}$
$e^{-at} \cos(bt) \mathcal{U}(t) \quad (a \in \mathbb{C}, \Re(a) > 0, b \in \mathbb{R})$	$\frac{a + i\omega}{(a + i\omega)^2 + b^2}$
$e^{-at^2} \quad (a > 0)$	$\sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}$